

Assignment 5: Congestion Control

Due date: Sunday, 31 May 2026 at 22:00

This is an individual assignment. You may discuss it with others, but your code and documentation must be written on your own. In particular, do not use automated code-generation tools such as ChatGPT or GitHub copilot.

This is a continuation of the development of a reliable transport protocol intended to offer a one-way, reliable and efficient stream communication system. For this assignment, you are required to extend the protocol by adding a congestion control mechanism. In particular, you must implement the Reno congestion control algorithm. The archive available on iCorsi contains a new sender and receiver implementation, `sender_cc.c` and `receiver_cc.c` respectively. For this assignment, you must modify the `sender_cc.c` implementation. `sender_cc.c` starts from the implementation seen in class with fast retransmission and RTT estimation. You should modify it such that the congestion window is not fixed but rather dynamically updated as a reaction to congestion as indicated by events such as timeouts, duplicate acknowledgments and in-sequence acknowledgments. You can assume a maximum window size of 1024 packets.

Submission Instructions

You must write your code in the `sender_cc.c` source file provided within the source package available on iCorsi.

Submit those completed files through the iCorsi system. Do not submit other files. Add comments to your code to explain sections of the code that might not be clear. You must also add comments at the beginning of the source file to properly acknowledge any and all external sources of information you may have used, including code, suggestions, and comments from other students. If your implementation has limitations and errors you are aware of (and were unable to fix), then list those as well in the initial comments.

You may use an integrated development environment (IDE) of your choice. However, *do not submit any IDE-specific file*, such as project description files.